

MORADABAD INSTITUTE OF TECHNOLOGY
COMPUTER SCIENCE & ENGINEERING
CLASS TEST-2 (3rd SEMESTER)

Session 2025-26

Subject: Digital Electronics
MM: 20
Branch: CSE

Sub Code: BOE310
Duration: 1 HR
Sec: A, B, C

Q.NO.	1	2	3	4	5	6
CO	CO5	CO4	CO5	CO3	CO3	CO5
Bloom's Level	K1	K2	K3	K2	K2	K3

CO3: Analyze and design of Sequential logic circuits with their applications.

CO4: Implement the Design procedure of Synchronous & Asynchronous Sequential Circuits.

CO5: Apply the concept of Digital Logic Families with circuit implementation

Note: 1) This paper contains two sections A and B.
2) All sections are compulsory.

Section A (08 Marks)

1. Define fan-in and fan-out in logic families. [2]
2. Explain hazards and its types. [3]
3. Implement the function $F1 = \sum (0, 1, 2, 5, 7)$ and $F2 = \sum (1, 2, 4, 6)$ using PLA. [3]

Section B (12 Marks)

4. With the help of logic diagram, explain the 4 bidirectional shift register. [4]
5. Explain the working and circuit of a modulo-5 counter. [4]
6. Design the basic circuit and truth table of TTL NAND gate. [4]

MORADABAD INSTITUTE OF TECHNOLOGY
COMPUTER SCIENCE & ENGINEERING
CLASS TEST-1 (3rd SEMESTER)

Session 2025-26

Subject: Digital Electronics
 MM: 20
 Branch: CSE

fair

Sub Code: BOE310
 Duration: 1 HR
 Sec: A, B, C

A-10
 B-11
 C-12
 D-13

Q.NO.	1	2	3	4	5	6
CO	1	1	1	3	2	2
Bloom's Level	K1	K2	K2	K3	K6	K6

CO1: Apply concepts of Digital Binary System and implementation of Gates.

CO2: Analyze and design of Combinational logic circuits

CO3: Analyze and design of Sequential logic circuits with their applications.

- Note: 1)** This paper contains two sections A and B.
2) All sections are compulsory.

105 138

Section A (08 Marks)

72 16 8 4 21
 1 0 1 1 10

1. Represent the unsigned decimal numbers 791 and 658 in BCD, and then show the steps necessary to form their sum. [2]
2. (i) Convert $(A5D8)_{16}$ into BCD number. [3×1=3]
 (ii) Convert $(46)_{10}$ to Gray Code.
 (iii) Represent -37 in 8-bit 2's complement.
3. Simplify using **K-map** and implement with **NAND** gates:
 $F(A, B, C, D) = \sum m(0, 1, 2, 5, 8, 9, 10, 13)$ [3]

Section B (12 Marks)

4. Derive the characteristic equation of SR flip flop with block diagram. [4]
5. Design a full adder by half adder and obtain the required expression for sum and carry. [4]
6. Design a combinational circuit that converts a four-bit binary number to a four-bit gray code. [4]

2 1 4 6
 2 2 3 0
 2 1 1 1
 2 5 1
 2 2 1
 1